

PU-I Biology — Chapter 3: Plant Kingdom

Worksheet: Classification Systems

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Name: _____

Section: _____

Date: _____

Section A — Fill in the Blanks

5 Questions · Classification Systems

1. The Artificial System of classification was given by _____.
2. The Natural System was proposed by Bentham and _____.
3. Cladistics groups organisms by _____ characters from a common ancestor.
4. Cytotaxonomy is based on _____ number, structure and behaviour.
5. Chemotaxonomy resolves confusion using the plant's _____ constituents.

Section B — One-word Answers

1. Who gave the Phylogenetic System of classification? _____
2. Which system is easily carried out using computers? _____
3. Name the two botanists behind the Natural System. _____
4. What term describes organisms grouped by shared evolutionary ancestry traits?

5. Name the five groups Plantae is now divided into. _____

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Section C — Match the Following

Column A	Column B
1. Artificial System	a. Shared derived characters
2. Natural System	b. Linnaeus
3. Phylogenetic System	c. Bentham & Hooker
4. Cladistics	d. Englar & Prantl
5. Cytotaxonomy	e. Chromosome data

Section D — True / False

1. The Artificial System gives equal importance to vegetative and sexual characters. _____
2. The Natural System is based only on leaf shape. _____
3. Numerical Taxonomy assigns numbers/codes to observable characters. _____
4. Cyanobacteria are still classified as algae under Plantae. _____
5. Chemotaxonomy uses chemical constituents of the plant. _____

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Section E — Short Answer

1. Why was the Artificial System of classification rejected?

2. Differentiate between the Natural System and the Phylogenetic System.

3. What is Cladistics based on?

Section F — HOTS (Higher Order Thinking)

1. Two plants look identical due to growing in the same shaded habitat. Explain which classification system(s) would correctly distinguish them and why.

2. Explain why numerical taxonomy is well-suited to computer-based analysis, using its basis of classification.

Teacher's Signature: _____ Date: _____

Teacher's Remarks:
